

```
In [1]: import pandas as pd
import numpy as np
import matplotlib
```

```
In [2]: df=pd.read_csv('Phase_1.csv')
df
```

Out[2]:

	Invoice (F)	StockCode	Description	Quantity	InvoiceDate (R)	Price	Customer ID	Country	TotalAmount (M)	Unnamed: 9	Unnamed: 10	Unnamed: 11
0	489434	85048	15CM CHRISTMAS GLASS BALL 20 LIGHTS	12	2009-01-12 07:45:00	6.95	13085	United Kingdom	83.4	NaN	R	Recency
1	489434	79323P	PINK CHERRY LIGHTS	12	2009-01-12 07:45:00	6.75	13085	United Kingdom	81.0	NaN	F	Frequency
2	489434	79323W	WHITE CHERRY LIGHTS	12	2009-01-12 07:45:00	6.75	13085	United Kingdom	81.0	NaN	M	Monetry
3	489434	22041	RECORD FRAME 7" SINGLE SIZE	48	2009-01-12 07:45:00	2.10	13085	United Kingdom	100.8	NaN	NaN	NaN
4	489434	21232	STRAWBERRY CERAMIC TRINKET BOX	24	2009-01-12 07:45:00	1.25	13085	United Kingdom	30.0	NaN	NaN	NaN
...	...	...	...	...	...	...	...	...	...	...	...	...
8335	490310	21551	CERAMIC BIRDHOUSE BUTTERFLY LARGE	6	2009-04-12 14:56:00	5.95	17151	United Kingdom	35.7	NaN	NaN	NaN
8336	490310	21550	CERAMIC BIRDHOUSE CRESTED TIT LARGE	12	2009-04-12 14:56:00	5.45	17151	United Kingdom	65.4	NaN	NaN	NaN
8337	490310	21552	CERAMIC BIRDHOUSE BLACK ROSE LARGE	12	2009-04-12 14:56:00	5.45	17151	United Kingdom	65.4	NaN	NaN	NaN
8338	490310	21554	CERAMIC BIRDHOUSE RED ROOF LARGE	12	2009-04-12 14:56:00	5.45	17151	United Kingdom	65.4	NaN	NaN	NaN
8339	490311	21232	STRAWBERRY CERAMIC TRINKET BOX	12	2009-04-12 15:11:00	1.25	16451	United Kingdom	15.0	NaN	NaN	NaN

8340 rows × 12 columns

```
In [29]: %store df #df Stored

Stored 'df' (DataFrame)
```

```
In [30]: %store -r
```

```
In [3]: df=df.loc[:, ~df.columns.str.contains('^Unnamed')]
df
```

Out[3]:

	Invoice (F)	StockCode	Description	Quantity	InvoiceDate (R)	Price	Customer ID	Country	TotalAmount (M)
0	489434	85048	15CM CHRISTMAS GLASS BALL 20 LIGHTS	12	2009-01-12 07:45:00	6.95	13085	United Kingdom	83.4
1	489434	79323P	PINK CHERRY LIGHTS	12	2009-01-12 07:45:00	6.75	13085	United Kingdom	81.0
2	489434	79323W	WHITE CHERRY LIGHTS	12	2009-01-12 07:45:00	6.75	13085	United Kingdom	81.0
3	489434	22041	RECORD FRAME 7" SINGLE SIZE	48	2009-01-12 07:45:00	2.10	13085	United Kingdom	100.8
4	489434	21232	STRAWBERRY CERAMIC TRINKET BOX	24	2009-01-12 07:45:00	1.25	13085	United Kingdom	30.0
...	...	...	...	...	...	...	...	...	...
8335	490310	21551	CERAMIC BIRDHOUSE BUTTERFLY LARGE	6	2009-04-12 14:56:00	5.95	17151	United Kingdom	35.7
8336	490310	21550	CERAMIC BIRDHOUSE CRESTED TIT LARGE	12	2009-04-12 14:56:00	5.45	17151	United Kingdom	65.4
8337	490310	21552	CERAMIC BIRDHOUSE BLACK ROSE LARGE	12	2009-04-12 14:56:00	5.45	17151	United Kingdom	65.4
8338	490310	21554	CERAMIC BIRDHOUSE RED ROOF LARGE	12	2009-04-12 14:56:00	5.45	17151	United Kingdom	65.4
8339	490311	21232	STRAWBERRY CERAMIC TRINKET BOX	12	2009-04-12 15:11:00	1.25	16451	United Kingdom	15.0

8340 rows × 9 columns

```
In [ ]: #Datatype Conversion
```

```
In [4]: df.dtypes
```

Out[4]: Invoice (F) int64
StockCode object
Description object
Quantity int64
InvoiceDate (R) object
Price float64
Customer ID int64
Country object
TotalAmount (M) float64
dtype: object

```
In [5]: df.columns
```

Out[5]: Index(['Invoice (F)', 'StockCode', 'Description', 'Quantity',
'InvoiceDate (R)', 'Price', 'Customer ID', 'Country',
'TotalAmount (M)'],
dtype='object')

```
In [6]: df=df[df['Quantity']>0].copy()
df
```

Out[6]:

	Invoice (F)	StockCode	Description	Quantity	InvoiceDate (R)	Price	Customer ID	Country	TotalAmount (M)
0	489434	85048	15CM CHRISTMAS GLASS BALL 20 LIGHTS	12	2009-01-12 07:45:00	6.95	13085	United Kingdom	83.4
1	489434	79323P	PINK CHERRY LIGHTS	12	2009-01-12 07:45:00	6.75	13085	United Kingdom	81.0
2	489434	79323W	WHITE CHERRY LIGHTS	12	2009-01-12 07:45:00	6.75	13085	United Kingdom	81.0
3	489434	22041	RECORD FRAME 7" SINGLE SIZE	48	2009-01-12 07:45:00	2.10	13085	United Kingdom	100.8
4	489434	21232	STRAWBERRY CERAMIC TRINKET BOX	24	2009-01-12 07:45:00	1.25	13085	United Kingdom	30.0
...	...	...	...	...	...	...	...	...	...
8335	490310	21551	CERAMIC BIRDHOUSE BUTTERFLY LARGE	6	2009-04-12 14:56:00	5.95	17151	United Kingdom	35.7
8336	490310	21550	CERAMIC BIRDHOUSE CRESTED TIT LARGE	12	2009-04-12 14:56:00	5.45	17151	United Kingdom	65.4
8337	490310	21552	CERAMIC BIRDHOUSE BLACK ROSE LARGE	12	2009-04-12 14:56:00	5.45	17151	United Kingdom	65.4
8338	490310	21554	CERAMIC BIRDHOUSE RED ROOF LARGE	12	2009-04-12 14:56:00	5.45	17151	United Kingdom	65.4
8339	490311	21232	STRAWBERRY CERAMIC TRINKET BOX	12	2009-04-12 15:11:00	1.25	16451	United Kingdom	15.0

8340 rows × 9 columns

```
In [7]: df['InvoiceDate (R)'] = pd.to_datetime(df['InvoiceDate (R)'])
df
```

Out[7]:

	Invoice (F)	StockCode	Description	Quantity	InvoiceDate (R)	Price	Customer ID	Country	TotalAmount (M)
0	489434	85048	15CM CHRISTMAS GLASS BALL 20 LIGHTS	12	2009-01-12 07:45:00	6.95	13085	United Kingdom	83.4
1	489434	79323P	PINK CHERRY LIGHTS	12	2009-01-12 07:45:00	6.75	13085	United Kingdom	81.0
2	489434	79323W	WHITE CHERRY LIGHTS	12	2009-01-12 07:45:00	6.75	13085	United Kingdom	81.0
3	489434	22041	RECORD FRAME 7" SINGLE SIZE	48	2009-01-12 07:45:00	2.10	13085	United Kingdom	100.8
4	489434	21232	STRAWBERRY CERAMIC TRINKET BOX	24	2009-01-12 07:45:00	1.25	13085	United Kingdom	30.0
...	...	...	...	...	...	...	...	...	...
8335	490310	21551	CERAMIC BIRDHOUSE BUTTERFLY LARGE	6	2009-04-12 14:56:00	5.95	17151	United Kingdom	35.7
8336	490310	21550	CERAMIC BIRDHOUSE CRESTED TIT LARGE	12	2009-04-12 14:56:00	5.45	17151	United Kingdom	65.4
8337	490310	21552	CERAMIC BIRDHOUSE BLACK ROSE LARGE	12	2009-04-12 14:56:00	5.45	17151	United Kingdom	65.4
8338	490310	21554	CERAMIC BIRDHOUSE RED ROOF LARGE	12	2009-04-12 14:56:00	5.45	17151	United Kingdom	65.4
8339	490311	21232	STRAWBERRY CERAMIC TRINKET BOX	12	2009-04-12 15:11:00	1.25	16451	United Kingdom	15.0

8340 rows × 9 columns

In [8]: df.dtypes

Out[8]: Invoice (F) int64
StockCode object
Description object
Quantity int64
InvoiceDate (R) datetime64[ns]
Price float64
Customer ID int64
Country object
TotalAmount (M) float64
dtype: object

In []: #Duplicates

In [9]: df = df.drop_duplicates()
df

Out[9]:

	Invoice (F)	StockCode	Description	Quantity	InvoiceDate (R)	Price	Customer ID	Country	TotalAmount (M)	
	0	489434	85048	15CM CHRISTMAS GLASS BALL 20 LIGHTS	12	2009-01-12 07:45:00	6.95	13085	United Kingdom	83.4
	1	489434	79323P	PINK CHERRY LIGHTS	12	2009-01-12 07:45:00	6.75	13085	United Kingdom	81.0
	2	489434	79323W	WHITE CHERRY LIGHTS	12	2009-01-12 07:45:00	6.75	13085	United Kingdom	81.0
	3	489434	22041	RECORD FRAME 7" SINGLE SIZE	48	2009-01-12 07:45:00	2.10	13085	United Kingdom	100.8
	4	489434	21232	STRAWBERRY CERAMIC TRINKET BOX	24	2009-01-12 07:45:00	1.25	13085	United Kingdom	30.0
	...	...	...	...	...	...	...	...	...	...
	8335	490310	21551	CERAMIC BIRDHOUSE BUTTERFLY LARGE	6	2009-04-12 14:56:00	5.95	17151	United Kingdom	35.7
	8336	490310	21550	CERAMIC BIRDHOUSE CRESTED TIT LARGE	12	2009-04-12 14:56:00	5.45	17151	United Kingdom	65.4
	8337	490310	21552	CERAMIC BIRDHOUSE BLACK ROSE LARGE	12	2009-04-12 14:56:00	5.45	17151	United Kingdom	65.4
	8338	490310	21554	CERAMIC BIRDHOUSE RED ROOF LARGE	12	2009-04-12 14:56:00	5.45	17151	United Kingdom	65.4
	8339	490311	21232	STRAWBERRY CERAMIC TRINKET BOX	12	2009-04-12 15:11:00	1.25	16451	United Kingdom	15.0

8213 rows × 9 columns

```
In [ ]: #Checking Outliers
```

```
In [ ]: #Verification of outliers in Quantity by Inter Quartile Range method (IQR)
```

```
In [10]: Q1=df['Quantity'].quantile(0.25)
Q3=df['Quantity'].quantile(0.75)
IQR=Q3-Q1
lower_bound = Q1 - 1.5*IQR
upper_bound = Q3 + 1.5*IQR
print("Lower Bound:", lower_bound)
print("Upper Bound;", upper_bound)
```

Lower Bound: -13.0

Upper Bound; 27.0

```
In [11]: df[(df['Quantity'] < lower_bound) |
(df['Quantity']>upper_bound)].shape
```

Out[11]: (521, 9)

```
In [ ]: #Verification of outliers in Total Amount by Inter Quartile Range method (IQR)
```

```
In [12]: Q1=df['TotalAmount (M)'].quantile(0.25)
Q3=df['TotalAmount (M)'].quantile(0.75)
IQR=Q3-Q1
```

```
lower_bound_amt = Q1 - 1.5*IQR
upper_bound_amt = Q3 + 1.5*IQR
print("Lower Bound:", lower_bound_amt)
print("Upper Bound;", upper_bound_amt)
```

Lower Bound: -18.25

Upper Bound; 42.15

```
In [13]: df[(df['TotalAmount (M)'] < lower_bound) |
           (df['TotalAmount (M)'] > upper_bound)].shape
```

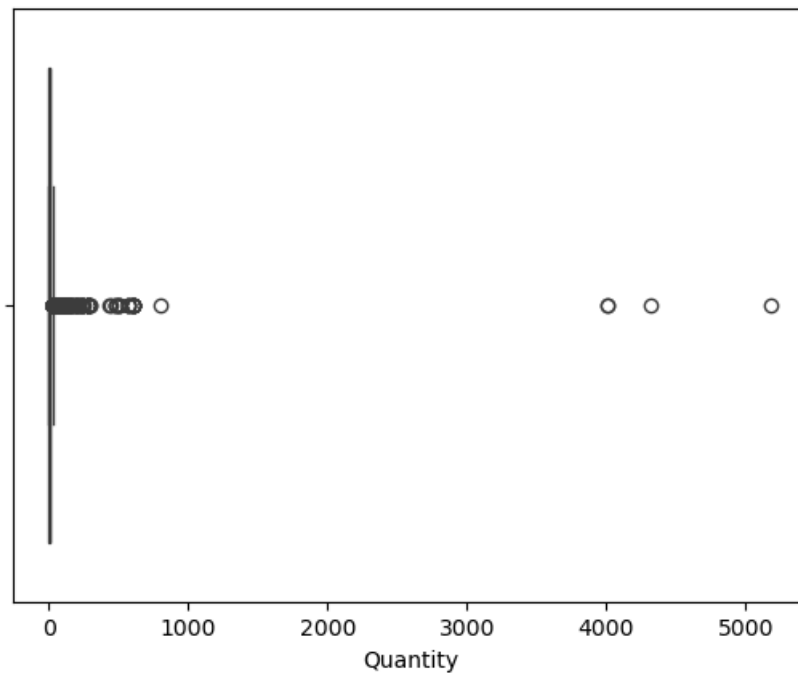
Out[13]: (1324, 9)

```
In [ ]: #Visual Illustration
```

```
In [14]: import seaborn as sns
import matplotlib.pyplot as plt
```

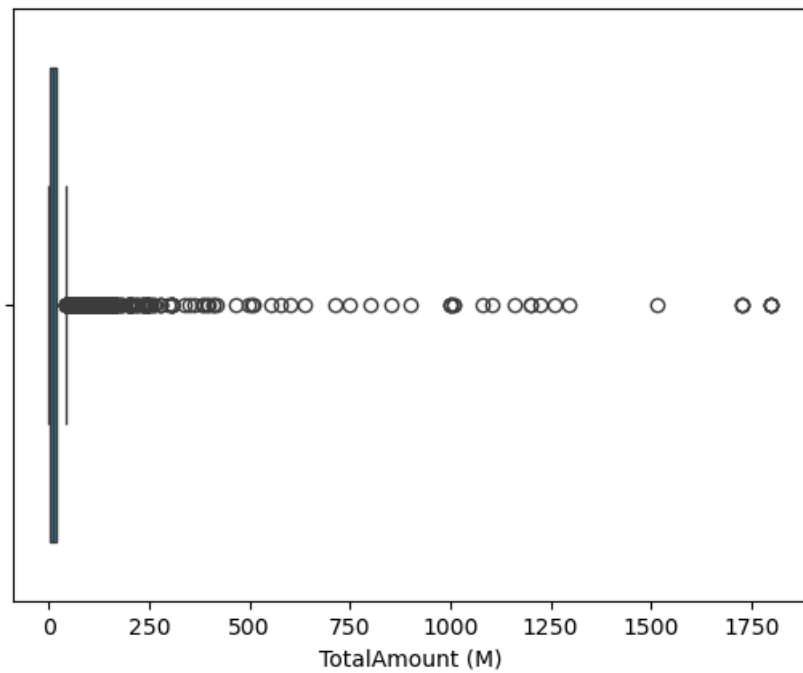
```
In [15]: sns.boxplot(x=df['Quantity'])
plt.show
```

Out[15]: <function matplotlib.pyplot.show(close=None, block=None)>



```
In [16]: sns.boxplot(x=df['TotalAmount (M)'])  
plt.show
```

```
Out[16]: <function matplotlib.pyplot.show(close=None, block=None)>
```



```
In [17]: outliers=df[df['Quantity']>upper_bound]  
len(outliers)
```

```
Out[17]: 521
```

```
In [18]: df.describe()
```

Out[18]:

	Invoice (F)	Quantity	InvoiceDate (R)	Price	Customer ID	TotalAmount (M)
count	8213.000000	8213.000000	8213	8213.000000	8213.000000	8213.000000
mean	489886.943504	14.220139	2009-02-22 20:11:54.652380416	3.044053	15497.382930	23.258079
min	489434.000000	1.000000	2009-01-12 07:45:00	0.060000	12362.000000	0.060000
25%	489654.000000	2.000000	2009-01-12 17:19:00	1.250000	14159.000000	4.400000
50%	489875.000000	4.000000	2009-02-12 15:41:00	2.100000	15581.000000	11.250000
75%	490100.000000	12.000000	2009-03-12 16:21:00	3.750000	16887.000000	19.500000
max	490311.000000	5184.000000	2009-04-12 15:11:00	141.000000	18229.000000	1800.000000
std	254.556837	103.560959	NaN	3.788703	1605.975105	80.097334

In [19]: df=df[df['Quantity']>0].copy()

In []: #Capping Quantity since the data has 6.2 percent outliers (large)

In [20]: df['Quantity']=df['Quantity'].clip(lower_bound, upper_bound)
df

Out[20]:

	Invoice (F)	StockCode	Description	Quantity	InvoiceDate (R)	Price	Customer ID	Country	TotalAmount (M)
0	489434	85048	15CM CHRISTMAS GLASS BALL 20 LIGHTS	12	2009-01-12 07:45:00	6.95	13085	United Kingdom	83.4
1	489434	79323P	PINK CHERRY LIGHTS	12	2009-01-12 07:45:00	6.75	13085	United Kingdom	81.0
2	489434	79323W	WHITE CHERRY LIGHTS	12	2009-01-12 07:45:00	6.75	13085	United Kingdom	81.0
3	489434	22041	RECORD FRAME 7" SINGLE SIZE	27	2009-01-12 07:45:00	2.10	13085	United Kingdom	100.8
4	489434	21232	STRAWBERRY CERAMIC TRINKET BOX	24	2009-01-12 07:45:00	1.25	13085	United Kingdom	30.0
...	...	...	...	...	...	...	...	...	...
8335	490310	21551	CERAMIC BIRDHOUSE BUTTERFLY LARGE	6	2009-04-12 14:56:00	5.95	17151	United Kingdom	35.7
8336	490310	21550	CERAMIC BIRDHOUSE CRESTED TIT LARGE	12	2009-04-12 14:56:00	5.45	17151	United Kingdom	65.4
8337	490310	21552	CERAMIC BIRDHOUSE BLACK ROSE LARGE	12	2009-04-12 14:56:00	5.45	17151	United Kingdom	65.4
8338	490310	21554	CERAMIC BIRDHOUSE RED ROOF LARGE	12	2009-04-12 14:56:00	5.45	17151	United Kingdom	65.4
8339	490311	21232	STRAWBERRY CERAMIC TRINKET BOX	12	2009-04-12 15:11:00	1.25	16451	United Kingdom	15.0

8213 rows × 9 columns


```
In [ ]: #Capping TotalAmount since the data has 15.8 percent outliers (large)
```

```
In [21]: df['TotalAmount (M)']=df['TotalAmount (M)'].clip(lower_bound, upper_bound)
df
```

Out[21]:

	Invoice (F)	StockCode	Description	Quantity	InvoiceDate (R)	Price	Customer ID	Country	TotalAmount (M)
0	489434	85048	15CM CHRISTMAS GLASS BALL 20 LIGHTS	12	2009-01-12 07:45:00	6.95	13085	United Kingdom	27.0
1	489434	79323P	PINK CHERRY LIGHTS	12	2009-01-12 07:45:00	6.75	13085	United Kingdom	27.0
2	489434	79323W	WHITE CHERRY LIGHTS	12	2009-01-12 07:45:00	6.75	13085	United Kingdom	27.0
3	489434	22041	RECORD FRAME 7" SINGLE SIZE	27	2009-01-12 07:45:00	2.10	13085	United Kingdom	27.0
4	489434	21232	STRAWBERRY CERAMIC TRINKET BOX	24	2009-01-12 07:45:00	1.25	13085	United Kingdom	27.0
...	...	...	...	...	...	...	...	...	...
8335	490310	21551	CERAMIC BIRDHOUSE BUTTERFLY LARGE	6	2009-04-12 14:56:00	5.95	17151	United Kingdom	27.0
8336	490310	21550	CERAMIC BIRDHOUSE CRESTED TIT LARGE	12	2009-04-12 14:56:00	5.45	17151	United Kingdom	27.0
8337	490310	21552	CERAMIC BIRDHOUSE BLACK ROSE LARGE	12	2009-04-12 14:56:00	5.45	17151	United Kingdom	27.0
8338	490310	21554	CERAMIC BIRDHOUSE RED ROOF LARGE	12	2009-04-12 14:56:00	5.45	17151	United Kingdom	27.0
8339	490311	21232	STRAWBERRY CERAMIC TRINKET BOX	12	2009-04-12 15:11:00	1.25	16451	United Kingdom	15.0

8213 rows × 9 columns

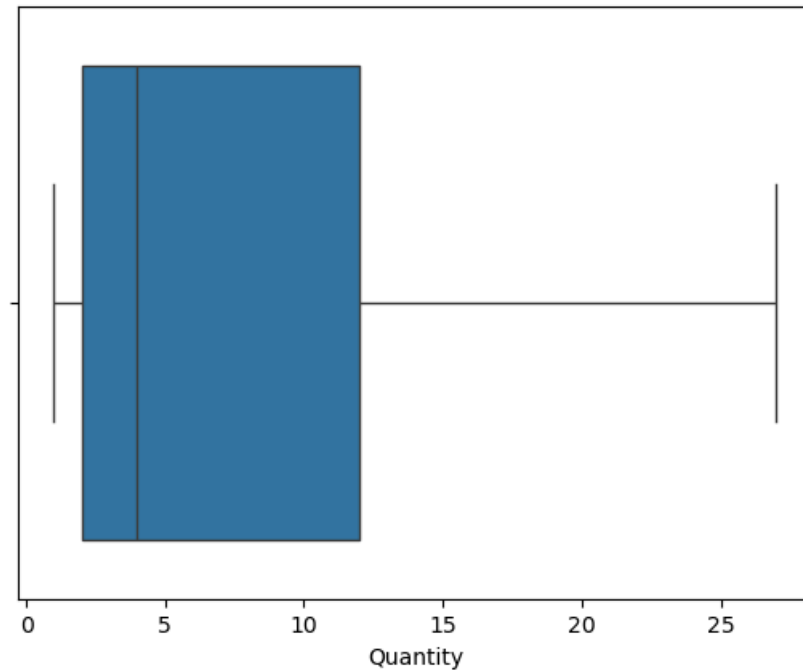
```
In [22]: df.describe()
```

Out[22]:

	Invoice (F)	Quantity	InvoiceDate (R)	Price	Customer ID	TotalAmount (M)
count	8213.000000	8213.000000	8213	8213.000000	8213.000000	8213.000000
mean	489886.943504	7.857178	2009-02-22 20:11:54.652380416	3.044053	15497.382930	12.652575
min	489434.000000	1.000000	2009-01-12 07:45:00	0.060000	12362.000000	0.060000
25%	489654.000000	2.000000	2009-01-12 17:19:00	1.250000	14159.000000	4.400000
50%	489875.000000	4.000000	2009-02-12 15:41:00	2.100000	15581.000000	11.250000
75%	490100.000000	12.000000	2009-03-12 16:21:00	3.750000	16887.000000	19.500000
max	490311.000000	27.000000	2009-04-12 15:11:00	141.000000	18229.000000	27.000000
std	254.556837	8.114631	NaN	3.788703	1605.975105	8.853517

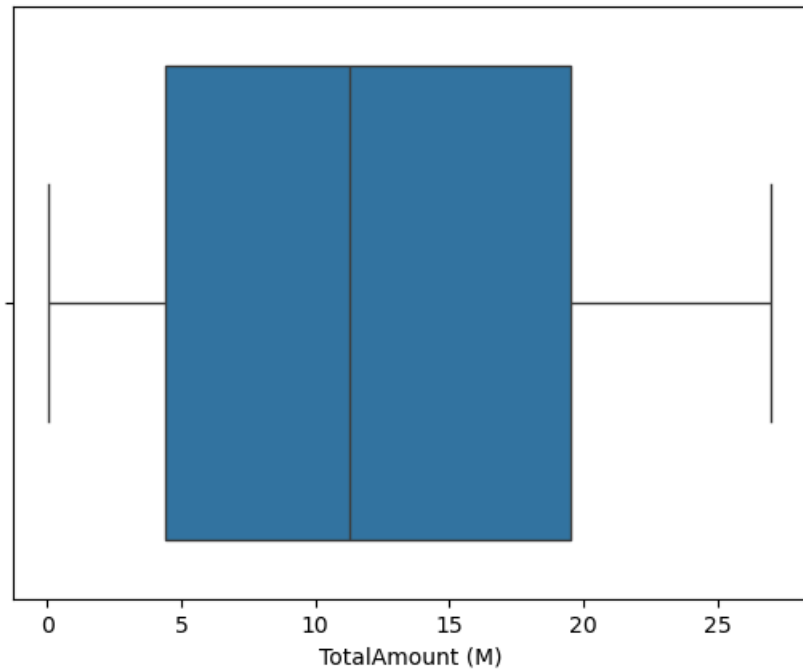
```
In [23]: sns.boxplot(x=df['Quantity'])  
plt.show
```

```
Out[23]: <function matplotlib.pyplot.show(close=None, block=None)>
```



```
In [24]: sns.boxplot(x=df['TotalAmount (M)'])  
plt.show
```

```
Out[24]: <function matplotlib.pyplot.show(close=None, block=None)>
```



```
In [ ]: #Snapshot Date
```

```
In [25]: df['InvoiceDate (R)'].min()  
df['InvoiceDate (R)'].max() #Date Range
```

```
Out[25]: Timestamp('2009-04-12 15:11:00')
```

```
In [26]: snapshot_date = df['InvoiceDate (R)'].max() + pd.Timedelta(days=1)  
snapshot_date
```

```
Out[26]: Timestamp('2009-04-13 15:11:00')
```

```
In [33]: df.columns = df.columns.str.strip()  
df
```

Out[33]:

	Invoice (F)	StockCode	Description	Quantity	InvoiceDate (R)	Price	Customer ID	Country	TotalAmount (M)
0	489434	85048	15CM CHRISTMAS GLASS BALL 20 LIGHTS	12	2009-01-12 07:45:00	6.95	13085	United Kingdom	27.0
1	489434	79323P	PINK CHERRY LIGHTS	12	2009-01-12 07:45:00	6.75	13085	United Kingdom	27.0
2	489434	79323W	WHITE CHERRY LIGHTS	12	2009-01-12 07:45:00	6.75	13085	United Kingdom	27.0
3	489434	22041	RECORD FRAME 7" SINGLE SIZE	27	2009-01-12 07:45:00	2.10	13085	United Kingdom	27.0
4	489434	21232	STRAWBERRY CERAMIC TRINKET BOX	24	2009-01-12 07:45:00	1.25	13085	United Kingdom	27.0
...	...	...	...	...	...	...	...	...	...
8335	490310	21551	CERAMIC BIRDHOUSE BUTTERFLY LARGE	6	2009-04-12 14:56:00	5.95	17151	United Kingdom	27.0
8336	490310	21550	CERAMIC BIRDHOUSE CRESTED TIT LARGE	12	2009-04-12 14:56:00	5.45	17151	United Kingdom	27.0
8337	490310	21552	CERAMIC BIRDHOUSE BLACK ROSE LARGE	12	2009-04-12 14:56:00	5.45	17151	United Kingdom	27.0
8338	490310	21554	CERAMIC BIRDHOUSE RED ROOF LARGE	12	2009-04-12 14:56:00	5.45	17151	United Kingdom	27.0
8339	490311	21232	STRAWBERRY CERAMIC TRINKET BOX	12	2009-04-12 15:11:00	1.25	16451	United Kingdom	15.0

8213 rows × 9 columns

```
In [28]: #df.to_excel("Cleaned_Data_2.xlsx", index=False)
```

```
In [ ]: #Stock Code Issue
```

```
In [ ]: #Checking Combo Products have single stock code.
```

```
In [32]: combo_df=df[df['Description'].str.contains(r'\+|&|/|\|,', regex=True, na=False)]
```

```
In [35]: combo_df
```

Out[35]:

	Invoice (F)	StockCode	Description	Quantity	InvoiceDate (R)	Price	Customer ID	Country	TotalAmount (M)	
	36	489437	21351	CINAMMON & ORANGE WREATH	2	2009-01-12 09:08:00	6.75	15362	United Kingdom	13.50
	37	489437	21352	EUCALYPTUS & PINECONE WREATH	2	2009-01-12 09:08:00	6.75	15362	United Kingdom	13.50
	51	489437	21912	VINTAGE SNAKES & LADDERS	4	2009-01-12 09:08:00	3.75	15362	United Kingdom	15.00
	61	489438	84031A	CHARLIE+LOLA RED HOT WATER BOTTLE	27	2009-01-12 09:24:00	3.00	18102	United Kingdom	27.00
	63	489438	84032A	CHARLIE+LOLA PINK HOT WATER BOTTLE	27	2009-01-12 09:24:00	1.90	18102	United Kingdom	27.00
	...	...	...	...	...	...	...	...	...	...
	8297	490305	72807C	SET/3 VANILLA SCENTED CANDLE IN BOX	8	2009-04-12 14:37:00	4.25	17243	United Kingdom	27.00
	8298	490305	72807A	SET/3 ROSE CANDLE IN JEWELLED BOX	4	2009-04-12 14:37:00	4.25	17243	United Kingdom	17.00
	8299	490305	72801F	S/4 FESTIVE RED DINNER CANDLES	12	2009-04-12 14:37:00	1.25	17243	United Kingdom	15.00
	8307	490305	84427B	BLACK GLASS TV DOUBLE P/FRA	3	2009-04-12 14:37:00	4.25	17243	United Kingdom	12.75
	8318	490305	21034	REX CASH+CARRY JUMBO SHOPPER	1	2009-04-12 14:37:00	0.95	17243	United Kingdom	0.95

822 rows × 9 columns

```
In [36]: combo_df['StockCode'].nunique() #Checking how many StockCode
```

Out[36]: 205

```
In [38]: combo_df.groupby('StockCode')['Description'].nunique().sort_values(ascending=False)
```

```
Out[38]: StockCode
16012      1
84656      1
84405A     1
84427B     1
84519A     1
..
22333      1
22334      1
22344      1
22345      1
90209C     1
Name: Description, Length: 205, dtype: int64
```

In []: